How should I investigate data quality?

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@roy ruddle

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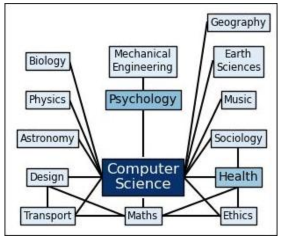
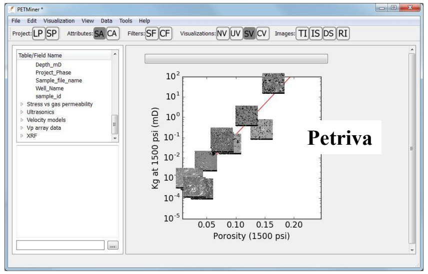
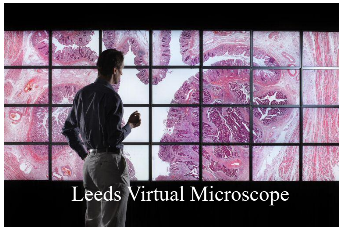


- Inter-disciplinary background
- Expert in data visualization and data quality
  - Hospital and GP data
  - Radio astronomy
  - Supermarkets & convenience stores
  - Adult social care
  - Petrophysics
  - Train doors!



Data quality
- Open source
- Method & PyPI package
- Training



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Three audiences

- Data scientists / analysts
  - Hands-on
- Business leaders / chief technical officers / dept. managers
  - The data scientists/analysts work for you
  - Wants: Efficient, profitable & high quality reputation
- Clients
  - Needs to be knowledgeable & informed
  - The project is being performed for <u>you</u>

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Overview

- Part 1
  - Introduction
  - 6-step Data Quality Method: What & when?
- Part 2
  - 6-step Data Quality Method: How? (open data example)
- Why? (benefits)
  - Save time
  - Reduce re-work
  - Improve results quality
- Q&A

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Part 1

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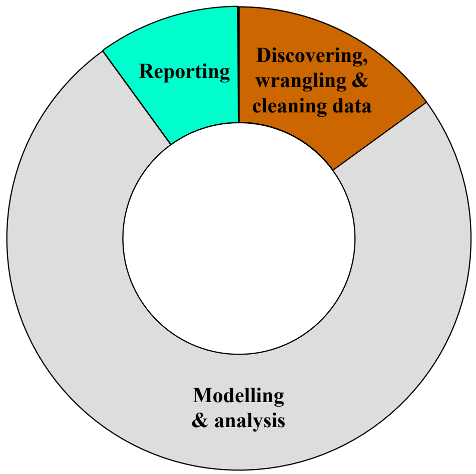
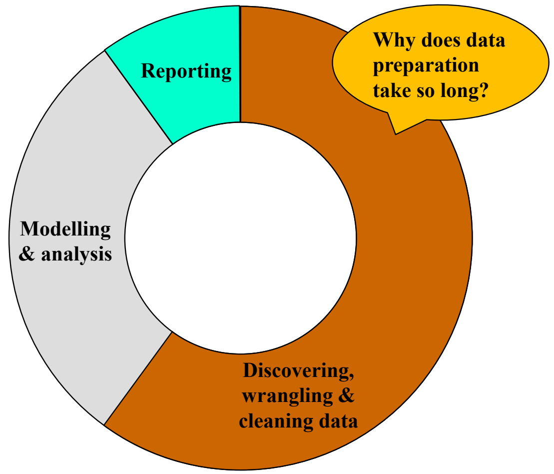
Workflow



Idealised data science project



Real project

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Lack of training / Ad hoc approaches

- Typical data scientist
  - Has little formal training in data quality
  - Cleans data in ad-hoc manner, following undocumented process
    - Only duplicate, missing value & missing record checks universally performed
  - Regrets not doing more
- Wants
  - To know what constitutes good practice
  - A rulebook (or formalised process) to follow

$\}$ 6-step Data Quality Method

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How has your data been “wrong”?

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How has your data been “wrong”?

Everything that can be wrong will be wrong, on some occasion … sooner than you think

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Solution

- **What?**
- **When?**
- **How?**
- **Why?**

6-step Data Quality Method
(https://github.com/royruddle/6-step-data-quality-method)

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What: A taxonomy of data quality

TAXONOMY OF DATA QUALITY TYPES AND CHARACTERISTICS (Level 1–3)

Type	Level 1	Level 2	Level 3
Accessibility [a,e,f,g]	Interpretability [f]		
Completeness [a,b,e,f,g] (also termed: Missingness [g], Omission [g], Presence [g], Sensitivity [g])	Missing data [c,d,e]	Missing record [d] (also termed: Missing tuple [c], Unit non-response [b])	
		Missing value [c,d]	Dummy entry [c] Item non-response [b] Semi-empty tuple [c] Zero value*
		Missing variable*	Completely missing variable* Empty column*
		Missing header*	Completely missing header* Missing column name*
	Duplicates [c,e]	Exact duplicates [c] Uniqueness violation [c,e] Duplicate header*	
	Coverage [b]	Appropriate amount of data [f]	Rate of recording [g]
		Concise representation [f]	
		Relevance [b,f]	Coding specificity [b] Relevant time intervals [b]
Accuracy [a,b,e,f,g] (also termed: Corrections made [g], Correctness [g], Errors [g], Incorrect [d], Positive predictive value [g])	Ambiguous data [c,e]	Abbreviations or imprecise/unusual coding [c]	
	Extreme [d]	Numeric outliers [d] Time-series outliers [d] Unusual category name* Special value*	
	Incorrect value [c,e] (also termed: Wrong data [c])	Coded wrongly or not conform to real entity [c] Domain violation [c,e] Embedded values [c] Erroneous entry [d] Extraneous data [d,e] Incorrect derived values [c] Measurement or recording error [b] Misfielded [c,d,e] Misspelling [c,e] Recording accuracy [b]	
		Validity [b,g]	Invalid substring [c]
	Misleading [g]	Objectivity [f]	
	Plausibility [g] (also termed: Believability [f,g], Implausible values [c], Trustworthiness [g])	Implausible range [c] Unexpected low/high values [c] Same value for too many records [c] Implausible changes of values over time [c] Pattern of value is unusual*	
	Syntax violation [c]	Data format [c] Wrong data type [c,d,e]	
Consistency [a,b,d,e,g] (also termed: Agreement [g], Concordance [g], Heterogeneity of representations [c, d, e], Reliability [b, g], Variation [g])	Heterogeneity of semantics [c] (also termed: Representational consistency [f])	Heterogeneity of aggregation/abstraction [c,e] Heterogeneity of measure units [c,d,e]	
		Inconsistent duplicates [c]	Approximate duplicates [c]
		Inconsistent spatial data [c] Information refers to different points in time [c,e] Misspelling (inconsistent) [d]	
		Naming conflicts [c]	Synonym/Homonym [c,e]
		Ordering [d] Violation of functional dependency [c,e]	
	Heterogeneity of syntaxes [c]	Different data formats [c] Different encoding formats [c,e] Different table structure [c] Different word orderings [c,e] Special characters [c,d,e]	
	References [c]	Referential integrity violation [c,e] Incorrect reference [c,e] Primary key violation [d] Circularity among tuples in a self-relationship [c]	

Dummy entry
Item non-response
Semi-empty tuple
Zero value

Approximate duplicates

Ruddle. (2023). Using well-known techniques to visualize characteristics of data quality. Proc. IVAPP.

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What?

- **69 characterisation and data quality tasks**
  - **Plain English**
  - **Each task articulated as question(s) to answer about your data**

Task	Question(s)
Example values	What does the data look like (print the first/last few rows)?
Range & variability	What is the minimum/maximum value? What are certain percentiles (e.g., 25th, median and 75th)?
Missing values	How many values are missing? Did you expect the variable to be complete?
Validity	Do any values contain characters or symbols that should not be present?
Violate assumptions?	What assumptions have you made about variables (are they true)?
Frequency measures	What is the distribution of the values? How many times does each value occur?

https://github.com/royruddle/6-step-data-quality-method/blob/main/downloads/tasks_and_questions.pdf

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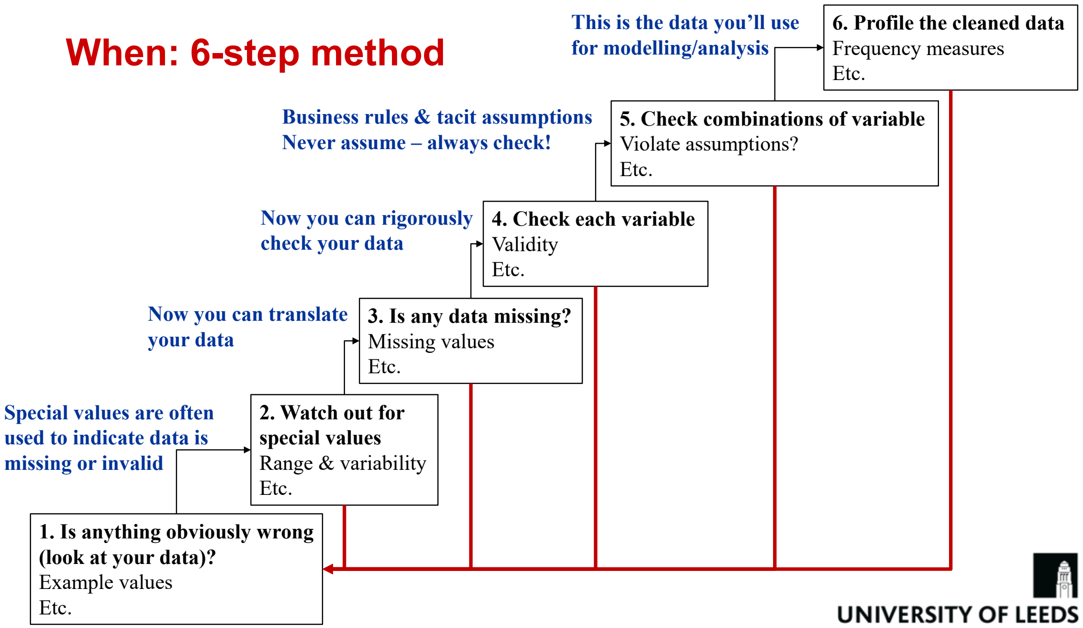
When: 6-step method



Task checklist (Excel; .ods)

Method step	Group	ID	Task
1	Examples	C-1	Example values
1	Documentation	C-13	Units
1	Encoding	C-2	File format
1	Documentation	DQ-1	Overall documentation
1	Duplicates	DQ-10	Duplicate header
1	Error	DQ-11	Incorrect column names
1	Format	DQ-12	Different table structure
1	Format	DQ-13	Different word orderings
1	Documentation	DQ-2	Documentation of variables
1	Documentation	DQ-3	Documentation of relationships
1	Missing data	DQ-4	Empty columns
1	Missing data	DQ-5	Empty rows
1	Missing data	DQ-6	Missing column names
1	Missing data	DQ-7	Missing header
1	Missing data	DQ-8	Missing records
1	Missing data	DQ-9	Missing variables
2	Missing data	C-11	Number of zero values
2	Distribution	C-17	Outliers
2	Distribution	C-18	Range and variability
2	Missing data	C-7	Number of infinite values
2	Missing data	C-9	Number of special values

https://github.com/royruddle/6-step-data-quality-method/blob/main/downloads/task_checklist.xlsx

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Part 2

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How?

- Calculations & visualizations with any software
- Worked example uses vizdataquality Python package
  - https://pypi.org/project/vizdataquality/
    - 16 example Jupyter notebooks (https://github.com/royruddle/vizdataquality)
  - Based on Pandas and Matplotlib
    - If necessary, customise visualizations with **kwargs
  - Clean your data & create a report at the same time
    - Output text, figures & tables as webpage, Latex or for Word

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Leeds parking fines

- Open data
  - https://datamillnorth.org/dataset/v8ggw/off-street-parking-fines
  - Quarterly datasets, Apr 2013 to Mar 2022
- Example dataset: Jan – Mar 2019
  - See vizdataquality "Workflow (parking fines)" notebook

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Step 1: Is anything obviously wrong?

- Look at your data
- Read any documentation (a sign of good quality data)

Data files that contain every fine issued for vehicles not having a valid parking ticket whilst in car parks. The data contains date, time, location, fine issued, discount available, and total paid.

Definitions

- Issued - Date PCN was issued
- Fine - Initial full charge (does not include surcharges)
- Surcharges - If full payment not made within 56 days from the date the notice is issued the charge is further increased by 50% of the Initial charge. If the payment is still not made, within a further 14 days then a further charge may be applied if the case is registered at court.
- Charge level - H = High at £70.00/£35.00 discount. L = Low at £50.00/£25.00 discount.
- Discount - the charge if payment made within 14 days.
- Balance Amount outstanding at date of sample. A negative balance means an overpayment.

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Step 1: Look at your data

```
df = pd.read_csv('Quarter 4 201819.csv')
```

```
print(df.iloc[-1])
```

```
PCN                                         LS08541469
ISSUED                                      31/03/2019
LOCATION                        WEST STREET CP - CITY
CONTRAVENTION   83 WITHOUT DISPLAYING A VALID TICKET
FINE                                                50
Last Pay Date                               31/03/2019
Total Paid                                         0.0
Balance                                            0.0
Unnamed: 8                                         NaN
```

Date format is "day first"

What time?

Eh?

Illogical

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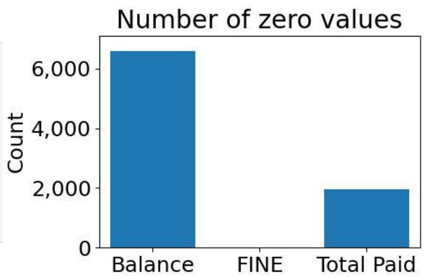
Step 2: Watch out for special values

- Watch out!
  - Certain values may have a special meaning (e.g., 999999)
  - Some systems use '0' as a default value

```
stats = vdqc.calc(df)
cols = ['Balance', 'FINE', 'Total Paid']
fig_kw = {'size_inches': (5, 3.25), 'constrained_layout': True}
ax_kw = {'title': 'Number of zero values', 'xlabel': None,
         'ylim': (0, num_rows), 'ylabel': 'Count'}
vdqp.scalar_bar(stats.loc[cols]['Number of zero values'],
                fig_kw=fig_kw, ax_kw=ax_kw)
```



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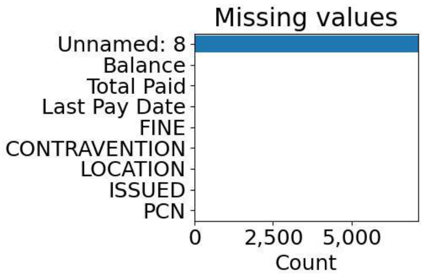
Step 3: Is any data missing?

- **Missing values**
  - **Special values?**
  - **Zeros?**
  - **Invalid values?**
- **What is the extra "Unnamed: 8" variable?**

```
fig_kw = {'size_inches': (5, 3.25), 'constrained_layout': True}
ax_kw = {'title': 'Missing values', 'ylabel': None, 'xlim': (0, num_rows),
         'xlabel': 'Count'}
vdqp.scalar_bar(stats['Number of missing values'], vert=False,
                fig_kw=fig_kw, ax_kw=ax_kw)
```

- **Spatial coverage**
- **Temporal coverage**



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Step 4: Check each variable

- **What checks would you perform?**

Data type	Variable
Categorical	PCN
Categorical	CONTRAVENTION
Categorical	LOCATION
Numerical	FINE
Numerical	Total Paid
Numerical	Balance
Date	ISSUED
Date	Last Pay Date

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A subset of the Step 4 tasks

- **Uniqueness**
- **Date & time distributions**
- **Value lengths**

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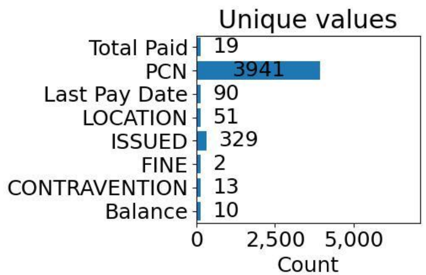
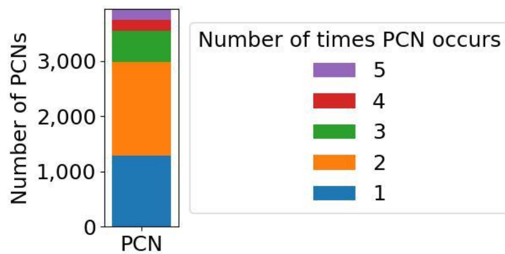
Step 4: Uniqueness

```
fig_kw = {'size_inches': (5, 3.25), 'constrained_layout': True}
ax_kw = {'title': 'Unique values', 'xlim': (0, num_rows), 'xlabel': 'Count',
         'ylabel': None}
vdqp.scalar_bar(stats['Number of unique values'], perceptual_threshold=0.02,
                vert=False, datalabels=True, fig_kw=fig_kw, ax_kw=ax_kw)
```





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Step 4: So how much was paid in total?

- **Some PCNs are paid in parts, not in one go**
  - **So you will get the wrong answer if you just sum the Total Paid**

```
# What is the sum of the Total Paid column?
print('Sum of the Total Paid column =', df['Total Paid'].sum())

# But what was the total amount paid?
print('Total actually paid =', df.groupby('PCN')['Total Paid'].max().sum())
```

```
Sum of the Total Paid column = 191106.02
Total actually paid = 106806.93
```

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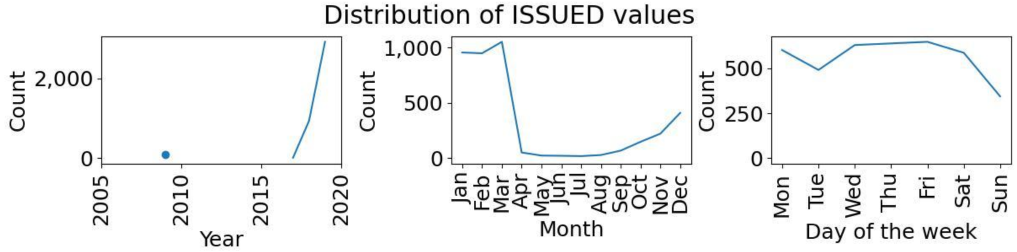
NB: Each PCN is only counted once!

Step 4: Dates

- **Check frequency distribution at multiple levels of detail**

```
fig_kw = {'size_inches': (12, 3), 'constrained_layout': True}
ax_kw = {'title': 'Distribution of ISSUED values'}
components = ['year', 'month', 'dayofweek']
xrot = [90, 90, 90]
vdqp.plotgrid('datetime distribution', df_noduplicates['ISSUED'], num_rows=1,
              xlabels_rotate=xrot, fig_kw=fig_kw, ax_kw=ax_kw, components=components)
```



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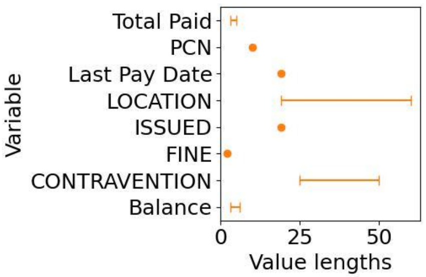
Step 4: Value lengths

- **How many characters are in the values of each variable?**
  - **Big ranges often indicate data quality issues**

```
fig_kw = {'size_inches': (3, 2)}
ax_kw = {'xlim': 0}
vdqp.dot_whisker(stats['Value lengths'].sort_index(), vert=False,
                 fig_kw=fig_kw, ax_kw=ax_kw)
```



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Step 4: Big value length ranges

```
cols = ['LOCATION', 'CONTRAVENTION']
df_examples = vdqc.get_value_lengths_examples(df[cols])

for column in df_examples.columns[1:]:
    print(column)
    for row in df_examples[['Examples', column]].itertuples(index=False):
        print(row[0].rjust(14) + ':', "'" + row[1] + "'")
```

```
LOCATION
Shortest value: 'CASTLE ST CP - CITY'
  Median value: 'KIRKSTALL LEISURE CENTRE - CP'
 Longest value: 'BARLEY HILL ROAD CP - GARFORTH                              '
CONTRAVENTION
Shortest value: '73 PARKED WITHOUT PAYMENT'
  Median value: '84 ADDITIONAL PAYMENT MADE TO EXTEND TIME'
 Longest value: '81 IN A RESTRICTED AREA IN A CAR PARK                       '
```

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Step 5: Check combinations of variables

- What assumptions have you made?
- Do any sets of values violate rules?

Data type	Variable
Categorical	PCN
Categorical	CONTRAVENTION
Categorical	LOCATION
Numerical	FINE
Numerical	Total Paid
Numerical	Balance
Date	ISSUED
Date	Last Pay Date

Step 5: Violate assumptions

- Why?

Fine – Total Paid – Balance (£)	Count
-116	3
-100	5
-56	4
-50	24
-43	18
-35	28
-33	729
-25	332
-10	1
0	785
25	3352
35	215
49.06	1
49.1	1
50	1559
70	51

Step 6: Profile the cleaned data

Data quality report

- Three formats
  - Text & images (manually import into Word)
  - Latex
  - HTML

Why?

- Do more in less time (know what to do; the tasks & questions)
- Reduce time & cost (by avoiding re-work)
- Understand any limitations of the data
- Correct your assumptions
- Improve your data

Improve results

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Summary

- Publicly available resources to help you rigorously investigate data quality and profile your data
  - 6-step Data Quality Method
  - Python package: vizdataquality
- One day CPD course
- YouTube "how to" videos (coming soon)
- Open data repository roll-out

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Questions

6-step Data Quality Method

https://github.com/royruddle/6-step-data-quality-method

vizdataquality

https://pypi.org/project/vizdataquality/

Slides, see https://github.com/royruddle/tutorials-and-talks/blob/main/README.md#talks-about-data-quality

Courses

Birmingham, London, Manchester, Edinburgh:
https://store.leeds.ac.uk/conferences-and-events/faculty-of-medicine-health/lida-leeds-institute-for-data-analytics

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